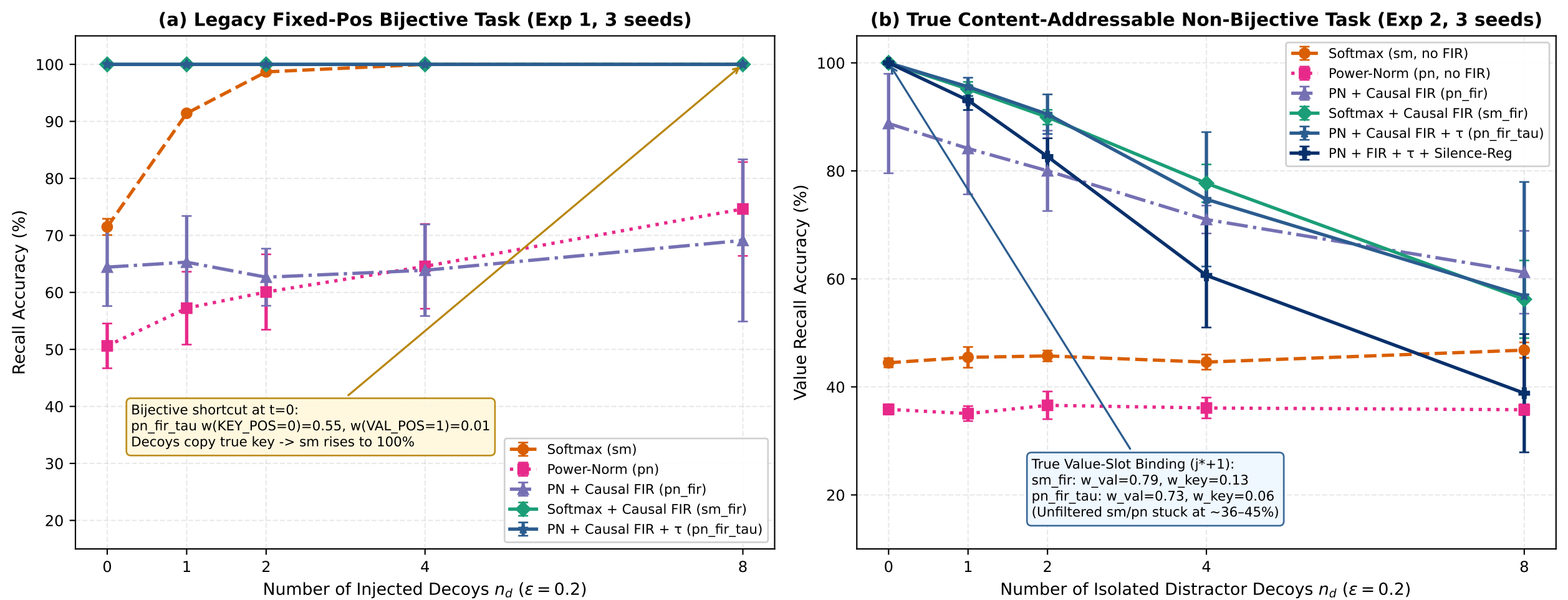
(a) Legacy Fixed-Pos Bijective Task (Exp 1, 3 seeds)
Recall Accuracy (%)
Number of Injected Decoys n_d ($\varepsilon = 0.2$)
Bijective shortcut at t=0:
pn_fir_tau w(KEY_POS=0)=0.55, w(VAL_POS=1)=0.01
Decoys copy true key -> sm rises to 100%
Softmax (sm)
Power-Norm (pn)
PN + Causal FIR (pn_fir)
Softmax + Causal FIR (sm_fir)
PN + Causal FIR + τ (pn_fir_tau)
(b) True Content-Addressable Non-Bijective Task (Exp 2, 3 seeds)
Value Recall Accuracy (%)
Number of Isolated Distractor Decoys n_d ($\varepsilon = 0.2$)
Softmax (sm, no FIR)
Power-Norm (pn, no FIR)
PN + Causal FIR (pn_fir)
Softmax + Causal FIR (sm_fir)
PN + Causal FIR + τ (pn_fir_tau)
PN + FIR + τ + Silence-Reg
True Value-Slot Binding (j*+1):
sm_fir: w_val=0.79, w_key=0.13
pn_fir_tau: w_val=0.73, w_key=0.06
(Unfiltered sm/pn stuck at ~36–45%)